

I Like the Way You Talk: Exploring Speech Influence on Social Priming Conversation Between Virtual Humans

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ABSTRACT

Social priming in virtual reality (VR) refers to utilizing an intentional conversation exposure to shape users' perceptions of a virtual human (VH), including social presence, affective attraction, etc. In this paper, we aim to investigate the effect of speech on social priming in VR. Specifically, we explore three factors of audibility, intelligibility, and information content by executing a within-subjects experiment. Our experiment reveal that intelligibility and information content positively affect social priming in VR. With experiment results, we provide a practical guidelines for social priming and virtual speech design.

Index Terms: Human-centered computing-Human computer interaction (HCI)-Interaction paradigms-Virtual reality.

1 INTRODUCTION

In virtual reality (VR), a prevalent social priming technique involves exposing users to a conversation between VHS [6]. This strategy has the potential to effectively enhance users' affect and social presence toward the VH. Therefore, in VR design, social priming offers an approach to shape users' perceptions of VHS. However, prior works lack the analysis of effective speech factors on users' perceptions, which restrict the flexible design of social priming [5]. Following prior research, we investigated the most fundamental speech factors: Audibility, Intelligibility and Information Content [1]. These respectively refer to whether the speech can be heard, whether it can be understood, and whether it conveys substantive information. We aimed to explore how these speech factors affect users' perceptions in VR social priming.

To that end, we conducted a within-subjects study and a mixed-methods analysis. Specifically, participants were exposed to a conversation between the target VH and a virtual interlocutor. For speech factors, four conditions were devised: Inaudible Speech, Gibberish Speech, Uninformative Speech, and Backstory Speech. Their pairwise comparison demonstrated the effects of the factors. We analyzed eye gaze, subjective questionnaires and post-study interviews through statistical methods and thematic analysis.

Our mixed-methods analysis revealed that in social priming, Intelligibility positively impacted users' social presence and affect of the VH. Information Content had a positive impact on social presence, and Audibility had no impact on users' perception. Based on the thematic analysis, we conducted an in-depth analysis of the speech factors that impact perception. Based on our findings, we derived guidelines for the design of social priming in VR.

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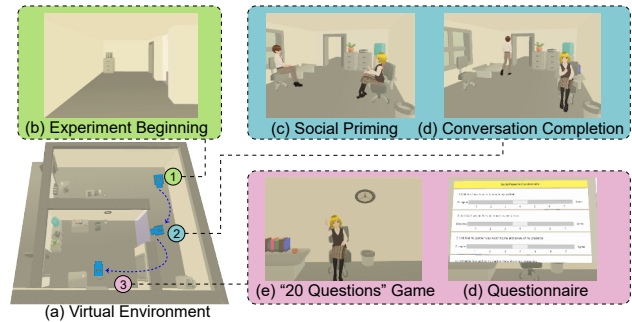


Figure 1: Figure (a) provides a top-down view of the virtual environment. The blue camera icon denotes the location of a participant, while the blue dashed arrows depict the trajectory of participant's movement.

Table 1: Speech Factors of Conditions

Factor \ Condition	INAUD	GIBB	UNINF	BACK
Audibility	No	Yes	Yes	Yes
Intelligibility	No	No	Yes	Yes
Information Content	Little	Little	Little	Much

2 VR EXPERIMENT

The environment and the procedure of VR experiment are shown in Fig. 1 (a). In each experiment, participants observed two VHS, the target VH and the virtual interlocutor. Participants were exposed to their conversation in different conditions, then completed a questionnaire about their perceptions of the target VH. During the experiment, all participants experienced the following conditions.

- **Inaudible Speech (INAUD):** Participants could see the conversation between the target VH and the virtual interlocutor but could not hear any voice. Factors: inaudible and uninformative.
- **Gibberish Speech (GIBB):** Participants could hear the conversation between the target VH and the virtual interlocutor, but the speech was gibberish. Factors: audible, unintelligible, and uninformative.
- **Uninformative Speech (UNINF):** The VHS exchanged farewells without sharing any contextual information. Factors: audible, intelligible, and uninformative.
- **Backstory Speech (BACK):** The target VH shared her hobby with the virtual interlocutor, who responded affirmatively. Factors: audible, intelligible, and informative.

The speech factors of each condition are shown in Tab. 1. To explore the effects of Audibility, we compared INAUD with GIBB. To explore the effects of Intelligibility, we compared GIBB with UNINF. To explore the effects of Information Content, we compared UNINF with BACK.

32 participants (9 female, 23 male, age 24-40, $M = 24.68$, $SD = 3.83$) were recruited for our study. Among them, 31 had computer gaming experience (from 1 to 7, $M = 6.1$), and 27 had used VR headsets before (from 1 to 7, $M = 4.8$). The study protocol was approved by the institutional review board of our university. All participants gave written informed consent before testing and were compensated for their participation.

Table 2: Summary of results. Statistical notations: F and X^2 denote ANOVA and Friedman tests for main effects; H indicates the Kruskal-Wallis test. Pairwise comparisons are reported using t (t-tests), Z (Wilcoxon), and U (Mann-Whitney U). p values indicate significance against $\alpha = 0.05$ (main effects) or $\alpha = 0.0125$ (pairwise comparisons). Significant data have been bolded.

Measures	Main Effect ($\alpha = 0.05$)	GIBB vs. INAUD	UNINF vs. GIBB	BACK vs. UNINF
Eye Gaze	$H(3) = 16.571, p < 0.001$	$U = 3482.5, p < 0.001$	$U = 4985.0, p = 0.012$	$U = 5477.5, p = 0.092$
Social Presence	$F(3, 29) = 23.169, p < 0.001$	$t(31) = -1.194, p = 0.242$	$t(31) = 3.078, p = 0.004$	$t(31) = 3.090, p = 0.004$
Affective Attraction	$X^2(3) = 30.709, p < 0.001$	$Z = -0.891, p = 0.373$	$Z = -3.403, p < 0.001$	$Z = -1.879, p = 0.06$
IOS	$X^2(3) = 21.637, p < 0.001$	$Z = -0.051, p = 0.959$	$Z = -2.866, p = 0.004$	$Z = -0.250, p = 0.803$

3 METHODS AND RESULTS

3.1 Quantitative Analysis

During observation of a turn-taking conversation, participants' eye gaze reflects attentional engagement [8]. We collected eye gaze during the social priming conversation to investigate participants' attention. We used time to first fixation (TTF) to quantify the delay between each turn transition of the speaker and the first fixation of eye gaze. Then we used Kruskal-Wallis tests to analyze differences and Mann-Whitney U tests for pairwise comparisons.

Drawing on previous work, we utilized four sets of classic questionnaires to evaluate participants' perceptions of target VH. The social presence questionnaire proposed by Biocca et al. [3] assessed how together participants felt with the VH. The following three were related to affect. The affective attraction questionnaire devised by Herbst et al. [7] evaluated participants' perceptions of the VHs. The pictorial inclusion of other in the self, proposed by Aron et al. [2], measured how close participants felt to the VHs.

We first employed Shapiro-Wilk tests and QQ plots to assess the normality of the data. For data that met the normality assumption, we utilized repeated measures ANOVA at the 5% significance level for analyzing differences. We used t-tests for pairwise comparisons. For data that did not meet the normality assumption, we utilized Friedman tests to analyze differences, and used Wilcoxon tests for pairwise comparisons.

The results of quantitative results is shown in Tab. 2. The comparison between GIBB and INAUD shows that there is no significant difference between them, which indicate that Audibility does not influence users' social presence and affect. The comparison between UNINF and GIBB shows that UNINF achieved better outcomes in all questionnaires. Therefore, we consider that Intelligibility have positive impacts on users' perceptions of VHs, including social presence and affect. The comparison between BACK and UNINF shows that users' social presence with the target VH was greater in BACK compared to UNINF. Data from eye-tracking suggests that speech without sound and speech cannot be understood fail to capture enough attention from participants.

3.2 Qualitative Analysis

We conducted a post-study interview to further collect participants' subjective opinions. For the analysis of post-study interviews, we used Braun and Clarke's thematic analysis approach [4]. Thematic analysis is a method for identifying, analyzing, and reporting themes within data, providing a rich and detailed account of qualitative information. We identified several themes based on the research questions and then merged, classified, and correlated these themes. Finally, we summarized the extracted themes and defined each of them.

We categorized all themes into two groups: Factor and Perception. Factor is defined as elements within speech during social priming that can exert an influence on participants' perceptions. Perception is defined as the users' perceptions of VHs that might be influenced by social priming. We identified 3 themes for Perception, including affect, social presence and sense of security. We also identified 3 themes for Factor, including factors of voice, factors of content and characteristics of VHs.

Twenty-three participants (71.9%) mentioned that factors of voice impacted their social presence toward the virtual human. Nine participants (27.3%) mentioned that factors of voice impacted

their affect of the virtual human. Eighteen participants (54.5%) indicated that factors of voice influenced their sense of security. Eight participants (24.2%) mentioned that factors of content enhanced social presence. Fourteen participants (42.4%) noted that factors of content improved their affect. Eighteen participants (54.5%) indicated that characteristics of VHs significantly influenced affect.

Based on the study findings, we have developed a series of guidelines for VR social priming design to direct VR designers in crafting the speech for shaping VHs and VR experiences. Social priming conversations should be audible and intelligible, thereby emphasizing the need of not disregarding the expenses associated with voice acting and script composition. To ensure that users perceive the target VH as an authentic individual who shares the same reality as the users, designers should prioritize Social Presence. Moreover, high Audibility is essential for attracting users' attention. Without auditory stimuli, even if the dialogue is positioned immediately in front of the user, it will not get adequate focus.

4 CONCLUSION

This study aimed to explore the effects of speech on social priming in VR. We experimentally explored that Intelligibility influenced users' social presence and affect of VHs, and that Information Content influenced users' social presence of VHs. Finally, we derived practical guidelines for the design of social priming in VR, supporting the precise shaping of users' perceptions of VHs.

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